

# Analysis of Salary Data

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# Agenda

Findings of linear regression modeling with tech salary data

- Data Description
- Regression Results
- Interpretation and Next Steps

# Data Description

The dataset contains 375 employee records with 6 features: Age, Gender, Education Level, Job Title, Years of Experience, and Salary.

- Missing Data: 2 rows had entirely missing values and were dropped. 1 row had an anomalous salary of \$350 (likely a data entry error) and was removed. Final clean dataset: 372 employees.
- Salary Range: \$30,000 to \$250,000, with a mean of \$100,847 and median of \$95,000.
- Education: 223 Bachelor's, 98 Master's, 51 PhD holders.
- Key correlations with salary:
  - - Years of Experience: strong positive correlation ( $r = 0.92$ )
  - - Education Level: PhD (\$143K avg) > Master's (\$108K avg) > Bachelor's (\$87K avg)
  - - Gender: Males (\$108K avg) earn more than Females (\$93K avg)

# Regression Results

Full Model: OLS regression with 10 features predicting Salary.

- R-squared: 0.917 (91.7% of salary variance explained)
- Significant variables ( $p < 0.05$ ) in reduced model:
  - - Years of Experience: +\$5,081 per year ( $p < 0.001$ )
  - - is\_director: +\$25,050 ( $p < 0.001$ )
  - - is\_senior: +\$14,350 ( $p < 0.001$ )
  - - is\_manager: +\$4,175 ( $p = 0.021$ )
  - - is\_male: +\$7,849 ( $p < 0.001$ )
  - - Master's degree: +\$13,970 vs Bachelor's ( $p < 0.001$ )
  - - PhD: +\$23,140 vs Bachelor's ( $p < 0.001$ )
- Removed (not significant): is\_junior ( $p=0.089$ ), is\_analyst ( $p=0.846$ ), is\_engineer ( $p=0.501$ )
- Reduced Model: R-squared = 0.916, Adj. R-squared = 0.914

# Interpretation and Next Steps

Variable 1: Years of Experience

- - Each additional year of experience increases salary by \$5,081
- - 95% CI: (\$4,737, \$5,426)
- - We are 95% confident salary increases between \$4,737 and \$5,426 per additional year
  
- Variable 2: Education Level (PhD vs Bachelor's)
- - A PhD holder earns \$23,140 more than a Bachelor's holder
- - 95% CI: (\$17,800, \$28,500)
- - We are 95% confident the true difference is between \$17,800 and \$28,500
  
- Next Steps:
- - Consider log-transforming salary to address right-skewed residuals
- - Investigate interaction effects (e.g., experience x education)
- - Address mild heteroscedasticity with robust standard errors